Chapter 10, Problem 33P

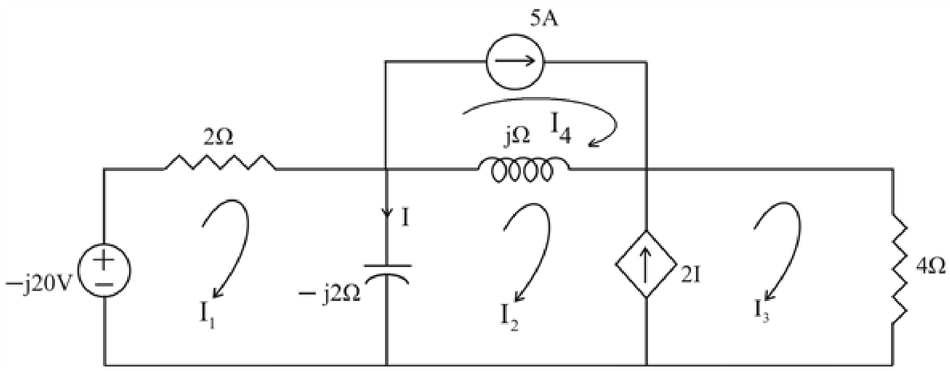
Problem

Compute **I** in Prob. 10.15 using mesh analysis.

Step-by-step solution

Step 1 of 4

Considering the circuit below:



Step 2 of 4

For Mesh 1,

$j20 + (2 - j2)I_1 + j2I_2 = 0$

$(1 - j)I_1 + jI_2 = -j10$ .... (1)

For the Supermesh,

$(j - j2)I_2 + j2I_1 + 4I_3 - jI_4 = 0$ .... (2)

Also,

$I_3 - I_2 = 2I$

$I_3 - I_2 = 2(I_1 - I_2)$

$I_3 = 2I_1 - I_2$ .... (3)

For Mesh 4,

$I_4 = 5\text{ A}$ .... (4)

Step 3 of 4

Substituting (3) & (4) into (2),

$$(8+j2)I_1-(4+j)I_2=j5$$

Putting (1) & (5) in matrix form,

$$\begin{bmatrix} 1-j & j \\ 8+j2 & -4-j \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix}=\begin{bmatrix} -j10 \\ j5 \end{bmatrix}$$

$$\Delta=-3-j5$$

$$\Delta_1=-5+j40$$

$$\Delta_2=-15+j85$$

Step 4 of 4

Therefore,

$$I=I_1-I_2$$

$$I=\frac{\Delta_1-\Delta_2}{\Delta}$$

$$I=\frac{10-j45}{-3-j5}$$

$$\therefore \boxed{I=7.906\angle 43.49^\circ \ \text{A}}$$